Lab Report 06

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1. Objective

The objective of this lab is to understand and implement the concept of loops in C++, especially for loop, nested for loop, and basic pattern generation.

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2. Apparatus / Software

- Computer System
- C++ Compiler (Code::Blocks / Dev C++ / VS Code)

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3. Theory

The for loop is used to execute a block of code a specific number of times. It consists of initialization, condition, and update expression. Nested loops are loops inside another loop and are commonly used for patterns and tables.

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4. Lab Tasks Implementation

Task 1: Display numbers 1 to 10 using loop

```
#include <iostream>
using namespace std;

int main() {

    // counter variable initialize kiya gaya hai
    int i = 1;

    // while loop ke zariye 1 se 10 tak numbers print kiye ja rahe hain
    while (i <= 10) {
        cout << i << endl;
        i++;
    }
```

```
    return 0;
}
```

Task 2: Multiplication Table (1 to 10)

```
#include <iostream>
using namespace std;

int main() {

    // outer loop rows ke liye use ho raha hai
    for (int i = 1; i <= 10; i++) {

        // inner loop columns ke liye use ho raha hai
        for (int j = 1; j <= 10; j++) {

            cout << i * j << "\t";
        }

        cout << endl;
    }

    return 0;
}
```

Task 3: Sum using for loop

```
#include <iostream>
using namespace std;

int main() {

    int n;
    int sum = 0;

    // user se positive integer input liya ja raha hai
    cin >> n;

    // 1 se n tak numbers ka sum calculate kiya ja raha hai
    for (int i = 1; i <= n; i++) {
        sum = sum + i;
    }

    // final sum display kiya ja raha hai
    cout << sum << endl;

    return 0;
}
```

Task 4: Nested Pattern Printing (Triangle)

```
#include <iostream>
using namespace std;

int main() {

    int rows;

    // user se rows ki number input li ja rahi hai
    cin >> rows;

    // nested loop ke zariye triangle pattern print kiya ja raha hai
    for (int i = 1; i <= rows; i++) {

        for (int j = 1; j <= i; j++) {
            cout << "*";
        }

        cout << endl;
    }

    return 0;
}
```

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5. Conclusion

In this lab, the concept of for loop and nested loops was implemented successfully. Different programs were developed for iteration, multiplication table generation, summation, and pattern printing.